

Phase I/II Operating Characteristics

New non-TITE alpha = 0.9 versus U-BOIN and BOIN12 | scenarios 1-38

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
OBD selection (%)						
New non-TITE one-stage	10.9	1.4	0.3	0	0	87.4
New non-TITE 2-stage highest utility	11.1	1.2	0.3	0	0	87.4
New non-TITE 2-stage top-2 randomized	10.5	2.1	0.1	0	0	87.3
U-BOIN	15.6	17.4	20.6	18.7	13.7	14
BOIN12	17.5	12	4.8	1.4	0.9	63.4
Mean administrations by dose						
New non-TITE one-stage	12	2.6	0.8	0.2	0	
New non-TITE 2-stage highest utility	11.9	2.6	0.9	0.2	0	
New non-TITE 2-stage top-2 randomized	11.9	2.7	0.9	0.2	0	
U-BOIN	9	5.4	4.7	3.9	3.7	
BOIN12	10	5.8	3.3	1.9	0.8	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
OBD selection (%)						
New non-TITE one-stage	36.2	26.5	30	4.6	0.1	2.6
New non-TITE 2-stage highest utility	45.4	23.5	22.5	5.3	0.6	2.7
New non-TITE 2-stage top-2 randomized	32.6	28.1	29	7	0.6	2.7
U-BOIN	11.4	13.5	41.5	21.8	10.6	1.2
BOIN12	27.8	24.6	32.7	9.9	1.3	3.7
Mean administrations by dose						
New non-TITE one-stage	13	8.2	7	1.3	0	
New non-TITE 2-stage highest utility	14.3	7	5.8	2	0.5	
New non-TITE 2-stage top-2 randomized	13	8	5.7	2.3	0.6	
U-BOIN	8.8	7.2	8.2	4	1.6	
BOIN12	8.5	7.6	8.6	3.6	1.1	

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New non-TITE alpha = 0.9 versus U-BOIN and BOIN12 | scenarios 1-38

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
OBD selection (%)						
New non-TITE one-stage	12	55.3	15.5	2.6	0.1	14.5
New non-TITE 2-stage highest utility	18.6	48.9	13.5	3.4	0.2	15.4
New non-TITE 2-stage top-2 randomized	10.5	53.3	15.9	4.3	0.5	15.5
U-BOIN	2.4	45	23.6	17.1	10.2	1.7
BOIN12	10.1	54	21.9	7.1	1.6	5.3
Mean administrations by dose						
New non-TITE one-stage	9.4	12.9	4.3	1.2	0.1	
New non-TITE 2-stage highest utility	10.6	10.5	4.4	1.8	0.4	
New non-TITE 2-stage top-2 randomized	10.6	9.3	5.3	2	0.5	
U-BOIN	7.2	10.6	6.5	3.7	1.7	
BOIN12	6.4	11.4	6.4	3.5	1.4	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
OBD selection (%)						
New non-TITE one-stage	10.6	8.6	23.8	48.9	5.8	2.3
New non-TITE 2-stage highest utility	11.7	11.2	29.7	38.6	6.5	2.3
New non-TITE 2-stage top-2 randomized	6.8	9.9	21.9	50.1	8.9	2.4
U-BOIN	3.8	4	18.8	46.4	27	0
BOIN12	12.6	14.6	26.9	35.6	7.1	3.2
Mean administrations by dose						
New non-TITE one-stage	4.7	5.7	7.9	9.8	1.5	
New non-TITE 2-stage highest utility	5	5.8	8.6	7.8	2.6	
New non-TITE 2-stage top-2 randomized	5	6.3	7.9	7.4	3.1	
U-BOIN	4.9	6.1	7.6	7.8	3.8	
BOIN12	5	5.7	7.2	8.8	3.1	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
OBD selection (%)						
New non-TITE one-stage	2.9	40	28.4	19.6	5.6	3.5
New non-TITE 2-stage highest utility	4.2	44	25.1	17.6	5.2	3.9
New non-TITE 2-stage top-2 randomized	3.3	39.6	27.4	20	5.9	3.8
U-BOIN	2.4	34.9	27.9	21.6	13.1	0.1
BOIN12	5.9	44.9	28.6	13.4	5.9	1.3
Mean administrations by dose						
New non-TITE one-stage	4.9	10.4	7.3	5	1.7	
New non-TITE 2-stage highest utility	5.1	10	7.1	4.9	2.1	
New non-TITE 2-stage top-2 randomized	5.3	8.6	7.3	5.5	2.6	
U-BOIN	5.3	8.5	7	5.6	3.8	
BOIN12	4.5	8.3	7.1	6	3.9	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
OBD selection (%)						
New non-TITE one-stage	67.4	6.6	0.8	0	0	25.2
New non-TITE 2-stage highest utility	67.3	5.7	0.6	0.1	0	26.3
New non-TITE 2-stage top-2 randomized	64.1	7.7	0.5	0.2	0	27.5
U-BOIN	21.5	17.1	15.8	19.3	15.1	11.2
BOIN12	59	17	5.4	1.8	0.1	16.7
Mean administrations by dose						
New non-TITE one-stage	21.5	3.5	0.6	0.1	0	
New non-TITE 2-stage highest utility	21.2	3.1	1	0.2	0	
New non-TITE 2-stage top-2 randomized	20.5	3.6	1.1	0.2	0	
U-BOIN	11.7	6.3	4.4	3.1	1.9	
BOIN12	15.3	6.5	3.3	1.6	0.6	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
OBD selection (%)						
New non-TITE one-stage	20.4	61.9	12.4	1.9	0	3.4
New non-TITE 2-stage highest utility	35.4	45.3	12.5	3	0.5	3.3
New non-TITE 2-stage top-2 randomized	18.5	56	18.3	3.7	0.5	3
U-BOIN	5.4	48.3	26.2	13.2	5.2	1.7
BOIN12	13.1	52.3	21.3	8.4	2.2	2.7
Mean administrations by dose						
New non-TITE one-stage	11.3	14.6	3	0.5	0	
New non-TITE 2-stage highest utility	13.2	9.6	4.3	1.8	0.5	
New non-TITE 2-stage top-2 randomized	12.5	9.1	5.1	2.2	0.6	
U-BOIN	7.9	10.8	6.5	3.2	1.3	
BOIN12	6.8	12.8	5.9	2.9	1.1	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
OBD selection (%)						
New non-TITE one-stage	27.1	2.3	0.1	0	0	70.5
New non-TITE 2-stage highest utility	27	1.9	0.4	0	0	70.7
New non-TITE 2-stage top-2 randomized	25.5	2.7	0.1	0	0	71.7
U-BOIN	12.6	15.7	15.2	18	23.6	14.9
BOIN12	29.9	15.1	4.2	1.3	0.2	49.3
Mean administrations by dose						
New non-TITE one-stage	15.2	2.4	0.5	0.1	0	
New non-TITE 2-stage highest utility	15.4	2.3	0.6	0.1	0	
New non-TITE 2-stage top-2 randomized	15.1	2.3	0.6	0.1	0	
U-BOIN	8.8	5	4.3	4.2	4.2	
BOIN12	11.7	5.9	3.1	1.8	0.7	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
OBD selection (%)						
New non-TITE one-stage	19.2	14.5	18.9	11.3	8.8	27.3
New non-TITE 2-stage highest utility	17.8	16.8	20.2	10.2	6.5	28.5
New non-TITE 2-stage top-2 randomized	14.9	16.1	22.1	11.9	7.4	27.6
U-BOIN	3.7	5.4	12.6	20.2	57.7	0.4
BOIN12	13.1	13.9	23.5	13	25.7	10.8
Mean administrations by dose						
New non-TITE one-stage	8.8	6.6	5.7	3.3	1.6	
New non-TITE 2-stage highest utility	8.9	6.7	6.1	2.8	1.5	
New non-TITE 2-stage top-2 randomized	8.9	6.7	5.6	3.2	1.6	
U-BOIN	6.6	5.7	6.3	5.4	6	
BOIN12	5.7	5.7	5.9	5	6.4	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
OBD selection (%)						
New non-TITE one-stage	10.3	23	5.6	1.1	0	60
New non-TITE 2-stage highest utility	9.3	23.6	5.3	0.6	0.2	61
New non-TITE 2-stage top-2 randomized	9	25	5.1	0.2	0.2	60.5
U-BOIN	1	31.6	24.8	19.6	18.6	4.4
BOIN12	8.8	42.9	16.2	5.4	1.5	25.2
Mean administrations by dose						
New non-TITE one-stage	10.8	7.3	2	0.4	0	
New non-TITE 2-stage highest utility	10.8	6.7	2.1	0.7	0.2	
New non-TITE 2-stage top-2 randomized	11.3	5.9	2.4	0.7	0.2	
U-BOIN	7.7	9	6.3	3.8	2.1	
BOIN12	7.3	10.1	5.2	2.6	1	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
OBD selection (%)						
New non-TITE one-stage	70.2	14.8	0.2	0.1	0	14.7
New non-TITE 2-stage highest utility	73.2	12	0.3	0.1	0	14.4
New non-TITE 2-stage top-2 randomized	69.4	15.1	0.2	0	0	15.3
U-BOIN	42.3	22.3	12.8	8.3	9.8	4.5
BOIN12	57.6	22.5	5.7	0.5	0	13.7
Mean administrations by dose						
New non-TITE one-stage	20.2	5.8	1.4	0.3	0	
New non-TITE 2-stage highest utility	21	5.1	1.4	0.2	0	
New non-TITE 2-stage top-2 randomized	20	5.8	1.7	0.3	0	
U-BOIN	11.6	6.7	4.7	3.3	2.7	
BOIN12	14.4	7.8	3.9	1.6	0.5	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
OBD selection (%)						
New non-TITE one-stage	19.2	14.3	11.4	41.3	10.7	3.1
New non-TITE 2-stage highest utility	21.7	19.8	11.9	33.9	8.9	3.8
New non-TITE 2-stage top-2 randomized	14.4	13.4	11.8	43.1	14.2	3.1
U-BOIN	6.8	10.1	6.4	39.1	37.6	0
BOIN12	16.6	23.8	15.1	32.3	9.9	2.3
Mean administrations by dose						
New non-TITE one-stage	5.3	6.5	6.3	8.8	2.7	
New non-TITE 2-stage highest utility	5.5	6.6	7.2	7.2	3.1	
New non-TITE 2-stage top-2 randomized	5.4	6.6	7	7.2	3.3	
U-BOIN	5.2	6.3	6.5	7.4	4.8	
BOIN12	5.2	6.2	5.9	8.2	4.4	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
OBD selection (%)						
New non-TITE one-stage	11.4	25.1	6	1.5	0.2	55.8
New non-TITE 2-stage highest utility	11	24.6	6.2	0.9	0.1	57.2
New non-TITE 2-stage top-2 randomized	10.1	27	5.3	0.9	0	56.7
U-BOIN	3.2	25.1	28.2	21	19.1	3.4
BOIN12	12.2	41.9	14.7	4.6	2	24.6
Mean administrations by dose						
New non-TITE one-stage	10.8	7.4	2.5	0.8	0.1	
New non-TITE 2-stage highest utility	10.8	7	2.6	0.8	0.2	
New non-TITE 2-stage top-2 randomized	11.2	6.4	2.7	0.9	0.2	
U-BOIN	7.8	7.5	6.5	4.4	3	
BOIN12	7.9	9.4	5.1	2.9	1.4	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
OBD selection (%)						
New non-TITE one-stage	26.5	16.7	14.5	24.2	5.7	12.4
New non-TITE 2-stage highest utility	29.2	20	12.7	20.5	5.4	12.2
New non-TITE 2-stage top-2 randomized	21.4	18.6	15.8	25.5	5.5	13.2
U-BOIN	11.4	8.4	10.1	35.8	34.1	0.2
BOIN12	23.8	21.2	16.7	22.8	8.6	6.9
Mean administrations by dose						
New non-TITE one-stage	7.1	7.3	6.6	5.7	1.9	
New non-TITE 2-stage highest utility	7.1	7.4	7.3	4.7	1.9	
New non-TITE 2-stage top-2 randomized	7.1	7.4	6.8	5	2	
U-BOIN	6.2	6.3	6.1	6.6	5	
BOIN12	6.2	6.7	6	6.6	4	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
OBD selection (%)						
New non-TITE one-stage	48.9	25.8	6.5	1.5	0.2	17.1
New non-TITE 2-stage highest utility	49.8	26	5.5	0.7	0.1	17.9
New non-TITE 2-stage top-2 randomized	43.5	28.4	9	1.9	0.1	17.1
U-BOIN	36.2	23.6	14.4	12.5	12.3	1
BOIN12	42.4	26.6	12.6	3.7	0.9	13.8
Mean administrations by dose						
New non-TITE one-stage	11.9	9.4	4.5	1.8	0.4	
New non-TITE 2-stage highest utility	12.3	9	4.6	1.6	0.4	
New non-TITE 2-stage top-2 randomized	11.5	8.7	5.3	2	0.4	
U-BOIN	9.2	7.5	5.6	4.2	3.4	
BOIN12	9.6	8.3	5.5	3.6	1.8	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
OBD selection (%)						
New non-TITE one-stage	25.3	29	20.4	16.5	8.8	0
New non-TITE 2-stage highest utility	26.5	32.3	16.8	16.6	7.8	0
New non-TITE 2-stage top-2 randomized	26.5	28.9	18.9	18.1	7.6	0
U-BOIN	28.1	26.8	21.1	16.3	7.7	0
BOIN12	35.1	27.5	19.7	12.1	5.6	0
Mean administrations by dose						
New non-TITE one-stage	7.2	8	6	5.3	3.6	
New non-TITE 2-stage highest utility	7.2	8.2	5.5	5.4	3.7	
New non-TITE 2-stage top-2 randomized	6.5	7.1	5.8	6.1	4.5	
U-BOIN	6.1	6.2	6.4	6.4	5.4	
BOIN12	4.9	5.7	6.7	7.4	5.3	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
OBD selection (%)						
New non-TITE one-stage	25.5	67.2	6.6	0.1	0	0.6
New non-TITE 2-stage highest utility	34.6	56.4	8.2	0.2	0	0.6
New non-TITE 2-stage top-2 randomized	20	65.7	13	0.7	0	0.6
U-BOIN	23.5	49.6	19.3	4.2	3.2	0.2
BOIN12	31.4	59	8.1	0.7	0	0.8
Mean administrations by dose						
New non-TITE one-stage	12.1	15.6	2.1	0.1	0	
New non-TITE 2-stage highest utility	12.1	12	4.6	1	0.1	
New non-TITE 2-stage top-2 randomized	12.3	10.9	5.4	1.2	0.1	
U-BOIN	10.1	11.9	5.9	1.7	0.4	
BOIN12	10.2	14	4.4	1.1	0.2	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
OBD selection (%)						
New non-TITE one-stage	10.4	10	10.2	20.5	48.8	0.1
New non-TITE 2-stage highest utility	13.5	12.4	13.4	20.3	40.2	0.2
New non-TITE 2-stage top-2 randomized	9.5	9.9	12.2	19.2	49.1	0.1
U-BOIN	7.1	13.1	16.3	15.4	48.1	0
BOIN12	16.1	20.1	21.6	14.4	27.6	0.2
Mean administrations by dose						
New non-TITE one-stage	4.1	5	5.6	6.7	8.5	
New non-TITE 2-stage highest utility	4.1	5.2	5.9	6.8	7.9	
New non-TITE 2-stage top-2 randomized	4.1	5.3	6.1	7.5	7.1	
U-BOIN	4.2	5.5	6.8	7.1	6.7	
BOIN12	4.3	5.2	5.8	6.6	8.1	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
OBD selection (%)						
New non-TITE one-stage	18.2	54.4	10.1	0.5	0	16.8
New non-TITE 2-stage highest utility	20.1	54.1	7.6	0.5	0	17.7
New non-TITE 2-stage top-2 randomized	14.3	57.1	10.1	0.7	0	17.8
U-BOIN	14	39.9	23	11.5	10.9	0.7
BOIN12	20.5	56.8	12.5	0.6	0	9.6
Mean administrations by dose						
New non-TITE one-stage	9.8	13	4.4	0.7	0	
New non-TITE 2-stage highest utility	10.4	12.5	4.1	0.9	0.1	
New non-TITE 2-stage top-2 randomized	10.9	10.8	4.9	1.1	0.1	
U-BOIN	8	10	6.3	3.5	2.1	
BOIN12	8.6	12	5.9	2	0.4	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
OBD selection (%)						
New non-TITE one-stage	9.5	54.4	25.1	5.9	0.3	4.8
New non-TITE 2-stage highest utility	9.9	57.8	22.8	4.3	0.2	5
New non-TITE 2-stage top-2 randomized	8.1	56.6	25.3	4.6	0.6	4.8
U-BOIN	8.5	47.5	18.6	12.4	13	0
BOIN12	11.3	62.1	18.8	4.9	0.1	2.8
Mean administrations by dose						
New non-TITE one-stage	5.8	12.7	7.3	2.9	0.5	
New non-TITE 2-stage highest utility	6	12.6	7.4	2.7	0.7	
New non-TITE 2-stage top-2 randomized	6.7	10.5	7.9	3.3	0.8	
U-BOIN	5.9	10.8	7	4	2.4	
BOIN12	6.2	10.7	7	4.1	1.7	

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	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
OBD selection (%)						
New non-TITE one-stage	42.1	0.1	0	0	0	57.8
New non-TITE 2-stage highest utility	40.3	0	0	0	0	59.7
New non-TITE 2-stage top-2 randomized	39.8	0.2	0	0	0	60
U-BOIN	51.8	8.3	5.7	5.7	3.7	24.8
BOIN12	60.5	0.5	0	0	0	39
Mean administrations by dose						
New non-TITE one-stage	20.1	0.6	0	0	0	
New non-TITE 2-stage highest utility	19	1.3	0.2	0	0	
New non-TITE 2-stage top-2 randomized	18.9	1.4	0.2	0	0	
U-BOIN	17.2	4	1.5	1	0.8	
BOIN12	20.7	2.5	0.4	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
OBD selection (%)						
New non-TITE one-stage	21.5	58.7	5	0.8	0	14
New non-TITE 2-stage highest utility	24.1	57	5	0.4	0	13.5
New non-TITE 2-stage top-2 randomized	16.3	58.5	8.6	0.2	0	16.4
U-BOIN	18.2	34.1	10.4	23.9	12.6	0.8
BOIN12	24.4	60.8	4	0.8	0	10
Mean administrations by dose						
New non-TITE one-stage	9.8	13.8	4.1	0.8	0	
New non-TITE 2-stage highest utility	10.1	13.8	4	0.6	0	
New non-TITE 2-stage top-2 randomized	11	11.9	4.7	0.7	0	
U-BOIN	8.2	9.8	5	4.5	2.5	
BOIN12	9.8	12.4	4.6	2	0.2	

Phase I/II Operating Characteristics

New non-TITE alpha = 0.9 versus U-BOIN and BOIN12 | scenarios 1-38

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
OBD selection (%)						
New non-TITE one-stage	5.1	0.1	0	0	0	94.8
New non-TITE 2-stage highest utility	6	0.1	0	0	0	93.9
New non-TITE 2-stage top-2 randomized	5.7	0.1	0	0	0	94.2
U-BOIN	15.2	16.2	6	2.5	1.6	58.5
BOIN12	10.7	0.7	0	0	0	88.6
Mean administrations by dose						
New non-TITE one-stage	11.8	1.5	0.1	0	0	
New non-TITE 2-stage highest utility	11.8	1.5	0.1	0	0	
New non-TITE 2-stage top-2 randomized	11.8	1.6	0.1	0	0	
U-BOIN	10.6	5.6	3.5	2.7	2.6	
BOIN12	11.6	4.8	1.2	0.1	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
OBD selection (%)						
New non-TITE one-stage	0.4	0.8	3.1	66.9	28.8	0
New non-TITE 2-stage highest utility	1	0.9	5.8	57.2	35.1	0
New non-TITE 2-stage top-2 randomized	0.6	0.6	3	55.9	39.9	0
U-BOIN	1.5	3.2	8.2	60.1	27	0
BOIN12	1.7	3.3	6.7	65.7	22.6	0
Mean administrations by dose						
New non-TITE one-stage	3.4	3.4	4.1	14	5.2	
New non-TITE 2-stage highest utility	3.4	3.4	4.1	11.2	7.9	
New non-TITE 2-stage top-2 randomized	3.5	3.8	4.7	9.5	8.5	
U-BOIN	3.5	4	5.1	9.8	7.9	
BOIN12	3.2	3.4	4.7	12.9	5.8	

Phase I/II Operating Characteristics

New non-TITE alpha = 0.9 versus U-BOIN and BOIN12 | scenarios 1-38

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
OBD selection (%)						
New non-TITE one-stage	30	40.4	27.1	1.3	0	1.2
New non-TITE 2-stage highest utility	36.5	40.8	20.3	0.9	0	1.5
New non-TITE 2-stage top-2 randomized	31.2	36.5	28.6	2.2	0	1.5
U-BOIN	28.1	26.8	25.8	9.6	9.4	0.3
BOIN12	40.7	35.7	21.7	0.4	0	1.5
Mean administrations by dose						
New non-TITE one-stage	10.6	10.6	7.3	1.2	0	
New non-TITE 2-stage highest utility	11.1	10.6	6.5	1.5	0.1	
New non-TITE 2-stage top-2 randomized	10.9	10	7	1.7	0.2	
U-BOIN	9.4	9.3	7.2	2.8	1.2	
BOIN12	9.9	9.9	7.6	2.1	0.3	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
OBD selection (%)						
New non-TITE one-stage	25.5	32.9	22.4	1.6	0	17.6
New non-TITE 2-stage highest utility	30.4	32.8	17.2	1	0.1	18.5
New non-TITE 2-stage top-2 randomized	22.7	37	21	0.9	0	18.4
U-BOIN	14.4	20.7	39.5	12.9	10.9	1.6
BOIN12	22.8	35.1	32.9	1	0	8.2
Mean administrations by dose						
New non-TITE one-stage	11.3	9.3	5.7	1.3	0	
New non-TITE 2-stage highest utility	12.2	8.7	5.3	1.2	0.1	
New non-TITE 2-stage top-2 randomized	11.8	8.7	5.3	1.5	0.1	
U-BOIN	7.8	7.7	8.2	4.1	2	
BOIN12	8.5	9.4	8	2.9	0.4	

Phase I/II Operating Characteristics

New non-TITE alpha = 0.9 versus U-BOIN and BOIN12 | scenarios 1-38

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
OBD selection (%)						
New non-TITE one-stage	16.6	32.7	39.1	3.3	0	8.3
New non-TITE 2-stage highest utility	22.4	32.8	33.3	2.9	0	8.6
New non-TITE 2-stage top-2 randomized	17	32.5	36.3	5.2	0.2	8.8
U-BOIN	8.7	20.9	38.9	16.5	14.5	0.5
BOIN12	20.9	32.9	38	4.2	0	4
Mean administrations by dose						
New non-TITE one-stage	8.4	9.4	9	1.9	0.1	
New non-TITE 2-stage highest utility	9.4	9.2	7.9	2.1	0.4	
New non-TITE 2-stage top-2 randomized	9.3	9.2	7.5	2.5	0.4	
U-BOIN	6.9	8.1	8.8	4.3	1.8	
BOIN12	7.2	9	9.2	3.4	0.7	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
OBD selection (%)						
New non-TITE one-stage	13.4	25.7	1.4	0	0	59.5
New non-TITE 2-stage highest utility	14.1	24.1	1	0	0	60.8
New non-TITE 2-stage top-2 randomized	14.7	23	2.4	0	0	59.9
U-BOIN	7.2	46	17.1	13.9	11.8	4
BOIN12	12.7	51.5	5.6	0.3	0.1	29.8
Mean administrations by dose						
New non-TITE one-stage	11.5	7.9	1.8	0.2	0	
New non-TITE 2-stage highest utility	11.5	7.9	1.8	0.2	0	
New non-TITE 2-stage top-2 randomized	12.2	7	2	0.3	0	
U-BOIN	8.2	9.5	5.4	3.6	2.6	
BOIN12	9.1	11.7	4.2	1.2	0.2	

Phase I/II Operating Characteristics

New non-TITE alpha = 0.9 versus U-BOIN and BOIN12 | scenarios 1-38

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
OBD selection (%)						
New non-TITE one-stage	75.3	21	0.5	0.1	0	3.1
New non-TITE 2-stage highest utility	78.8	17.4	0.3	0	0	3.5
New non-TITE 2-stage top-2 randomized	71.1	23.4	1.7	0	0	3.8
U-BOIN	57.7	26.2	6.6	2.4	4.4	2.7
BOIN12	72.7	22.9	1.2	0	0	3.2
Mean administrations by dose						
New non-TITE one-stage	21.8	6.9	0.6	0	0	
New non-TITE 2-stage highest utility	20.8	6.5	1.8	0.2	0	
New non-TITE 2-stage top-2 randomized	19.5	7.6	2.1	0.2	0	
U-BOIN	16.2	8.5	3.2	1	0.5	
BOIN12	17.4	8.6	2.7	0.6	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
OBD selection (%)						
New non-TITE one-stage	15.6	16.9	19	41.4	6.6	0.5
New non-TITE 2-stage highest utility	20.3	21.8	20.5	32.9	4	0.5
New non-TITE 2-stage top-2 randomized	14.3	17.7	15.7	42.7	8.8	0.8
U-BOIN	13	15.8	16.7	37.7	16.8	0
BOIN12	23.9	23.6	21.2	28.2	2.4	0.7
Mean administrations by dose						
New non-TITE one-stage	6.1	6.9	6.3	8.6	2.1	
New non-TITE 2-stage highest utility	6.2	7	6.8	7.2	2.6	
New non-TITE 2-stage top-2 randomized	5.8	6.7	6.9	7.4	3.1	
U-BOIN	5.8	6.8	7.1	7.2	3.4	
BOIN12	5.3	6.1	6.6	8.6	3.3	

Phase I/II Operating Characteristics

New non-TITE alpha = 0.9 versus U-BOIN and BOIN12 | scenarios 1-38

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
OBD selection (%)						
New non-TITE one-stage	2.1	59.1	28.6	10	0.2	0
New non-TITE 2-stage highest utility	4.4	45	26.3	21.7	2.6	0
New non-TITE 2-stage top-2 randomized	2.3	40.9	28.6	23.9	4.3	0
U-BOIN	3.2	42.6	32.7	20	1.5	0
BOIN12	3.3	41.2	34.5	19.9	1	0.1
Mean administrations by dose						
New non-TITE one-stage	4.5	16.9	6.6	1.9	0.1	
New non-TITE 2-stage highest utility	4.5	9.3	6.9	6.4	2.9	
New non-TITE 2-stage top-2 randomized	4.5	8.1	7.2	7	3.2	
U-BOIN	4.4	7.7	7.8	7.4	2.9	
BOIN12	3.9	12.3	7.9	4.8	1.1	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
OBD selection (%)						
New non-TITE one-stage	8.7	12.5	40.4	30	8.3	0.1
New non-TITE 2-stage highest utility	8.1	16.6	40.3	28.6	6.2	0.2
New non-TITE 2-stage top-2 randomized	7.3	13.2	39.7	31.2	8.5	0.1
U-BOIN	8.5	16	45.7	17.4	12.4	0
BOIN12	14.1	21.4	46.4	13.3	4.6	0.2
Mean administrations by dose						
New non-TITE one-stage	4.3	5.8	9.4	7.7	2.8	
New non-TITE 2-stage highest utility	4.4	5.8	8.9	7.5	3.3	
New non-TITE 2-stage top-2 randomized	4.4	5.9	8.2	7.4	4.1	
U-BOIN	4.5	6.1	9.8	6.5	3.3	
BOIN12	4.4	5.9	9.2	6.7	3.8	

Phase I/II Operating Characteristics

New non-TITE alpha = 0.9 versus U-BOIN and BOIN12 | scenarios 1-38

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
OBD selection (%)						
New non-TITE one-stage	9.5	5.5	15.5	51.5	18	0
New non-TITE 2-stage highest utility	10.5	7.7	16.9	48.7	16.2	0
New non-TITE 2-stage top-2 randomized	5.5	4.6	12.9	55	22	0
U-BOIN	6.1	8.6	24.9	28.6	31.8	0
BOIN12	12.3	14.7	33.2	28.1	11.4	0.3
Mean administrations by dose						
New non-TITE one-stage	3.8	4.3	6.4	11.1	4.4	
New non-TITE 2-stage highest utility	3.7	4.1	6.5	10.1	5.6	
New non-TITE 2-stage top-2 randomized	3.7	4.6	6.9	9.1	5.7	
U-BOIN	3.8	5	8.8	7.8	4.6	
BOIN12	4.4	5.5	7.9	7.9	4.3	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
OBD selection (%)						
New non-TITE one-stage	17.5	17.6	45.3	12.9	1.6	5.1
New non-TITE 2-stage highest utility	17.3	18.5	47	10.5	1.1	5.6
New non-TITE 2-stage top-2 randomized	12.2	19.7	46.3	14.3	1.6	5.9
U-BOIN	9.1	11.4	26	26.5	27	0
BOIN12	20.5	22.2	37.8	11	3	5.5
Mean administrations by dose						
New non-TITE one-stage	5.8	8.1	10.3	4.5	0.8	
New non-TITE 2-stage highest utility	5.8	8.2	10.5	4	1.1	
New non-TITE 2-stage top-2 randomized	6.2	8.3	9.2	4.3	1.3	
U-BOIN	5.9	7.5	7.8	5.3	3.6	
BOIN12	6.6	7.8	8	4.9	2.4	

Phase I/II Operating Characteristics

New non-TITE alpha = 0.9 versus U-BOIN and BOIN12 | scenarios 1-38

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
OBD selection (%)						
New non-TITE one-stage	5.8	61.4	26.7	5.9	0.2	0
New non-TITE 2-stage highest utility	8.7	56.8	27.2	6.5	0.8	0
New non-TITE 2-stage top-2 randomized	5.6	53.5	28.6	11	1.2	0.1
U-BOIN	10.1	61.8	19.4	5.7	3	0
BOIN12	12.3	63.1	19.3	4.2	1.1	0
Mean administrations by dose						
New non-TITE one-stage	5.4	16.4	6.6	1.5	0.1	
New non-TITE 2-stage highest utility	5.2	12.2	8	3.5	1	
New non-TITE 2-stage top-2 randomized	5.8	10.1	8.7	4.3	1.2	
U-BOIN	5.8	13.2	7.5	2.7	0.8	
BOIN12	5.3	13.5	7.1	3	1.1	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
OBD selection (%)						
New non-TITE one-stage	17.8	36.2	7.6	1.6	0.3	36.5
New non-TITE 2-stage highest utility	16.6	36.6	7.9	1.6	0.2	37.1
New non-TITE 2-stage top-2 randomized	13	40.5	7.2	1.6	0.2	37.5
U-BOIN	7.8	18.8	15.2	24.1	33.2	0.9
BOIN12	17.6	43.5	13.5	4.9	1.9	18.6
Mean administrations by dose						
New non-TITE one-stage	9.9	10.3	3.9	1.2	0.2	
New non-TITE 2-stage highest utility	10.1	10.1	3.9	1.2	0.3	
New non-TITE 2-stage top-2 randomized	10.5	9.2	4.1	1.2	0.3	
U-BOIN	7.9	7.5	5.4	5	4	
BOIN12	8.4	10.1	4.9	3.2	1.6	

Phase I/II Operating Characteristics

New non-TITE alpha = 0.9 versus U-BOIN and BOIN12 | scenarios 1-38

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
OBD selection (%)						
New non-TITE one-stage	19.2	44.1	7.3	0.9	0.1	28.4
New non-TITE 2-stage highest utility	19.6	44.9	6.8	0.7	0	28
New non-TITE 2-stage top-2 randomized	12.5	50.1	7.3	0.7	0	29.4
U-BOIN	16.5	27.5	19.3	16.9	19.4	0.4
BOIN12	21	49.5	10	2.2	0.2	17.1
Mean administrations by dose						
New non-TITE one-stage	9.5	12.3	4.2	1	0.1	
New non-TITE 2-stage highest utility	9.6	12.3	4.2	1	0.2	
New non-TITE 2-stage top-2 randomized	10.1	11	4.6	1.1	0.2	
U-BOIN	8.4	8.3	5.5	4.3	3.5	
BOIN12	9.3	11	4.7	2.4	1	
Scenario 38						
DLT rate	0.05	0.1	0.12	0.15	0.3	
Efficacy rate	0.05	0.08	0.12	0.15	0.5	
OBD selection (%)						
New non-TITE one-stage	14.5	10.6	9.6	12.2	51.3	1.8
New non-TITE 2-stage highest utility	21.1	12.8	11.8	11.7	40.3	2.3
New non-TITE 2-stage top-2 randomized	12.1	10.6	11.1	11.3	52.6	2.3
U-BOIN	6.2	8.9	13	13.3	58.6	0
BOIN12	13.2	13.8	16.5	13.6	41.5	1.4
Mean administrations by dose						
New non-TITE one-stage	4.9	5.4	5.6	5.6	8.4	
New non-TITE 2-stage highest utility	5.2	5.6	6	5.8	7.2	
New non-TITE 2-stage top-2 randomized	5	5.7	6	6.5	6.5	
U-BOIN	5.2	5.5	5.6	6.3	7.8	
BOIN12	4.2	4.7	5.2	6.1	9.6	